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Linear Algebra

5Week [02 October - 08 October]

Quiz 5

Started on	2023-10-09 07:30
State	Finished
Completed on	2023-10-09 07:48
Time taken	17 mins 33 secs
Marks	4.50/5.00
Grade	1.80 out of 2.00 (90%)

Question 1

Correct

Mark 1.00 out of 1.00

Find $\begin{vmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{vmatrix}$, if $\varphi = 75^\circ$.

Select one:

- ☐ $-\frac{\sqrt{3}}{2}$
- ☐ 0
- ☐ $-\frac{2}{\sqrt{3}}$
- ☒ 1 ✓

Your answer is correct.

The correct answer is: 1

Question 2

Correct

Mark 1.00 out of 1.00

Solve the equation. $\begin{vmatrix} 2x+1 & 3 \\ x+5 & 2 \end{vmatrix} = 0$.

Select one:

- ☒ 13 ✓
- ☐ 14
- ☐ -10
- ☐ 0

Your answer is correct.

The correct answer is: 13

Question 3
Partially correct
Mark 0.50 out of 1.00

Find true statements.

Select one or more:

- ☐ If $\det A = 0$, then matrix is invertible.
- ☐ If matrix A has zero column, then $\det A = 0$.
- ☒ If matrix has two equal columns, then its determinant is zero. ✓
- ☒ If A is 3×4 matrix, then $\det A^T = \det A$. ✗

Your answer is partially correct.

You have correctly selected 1.

The correct answers are: If matrix A has zero column, then

$\det A = 0$. If matrix has two equal columns, then its determinant is zero.

Question 4
Correct
Mark 1.00 out of 1.00

Find $\begin{bmatrix} a & 0 & 3 & 5 \\ 0 & 0 & b & 2 \\ 1 & c & 2 & 3 \\ 0 & 0 & 0 & d \end{bmatrix}$, if $a = b = c = d = 2$.

Select one:

- ☐ 16
- ☒ -16 ✓
- ☐ 0
- ☐ 8

Your answer is correct.

The correct answer is: -16

Question 5
Correct
Mark 1.00 out of 1.00

Find $\begin{vmatrix} 10 & 5 + 2 \cos \alpha & \cos \alpha \\ 12 & 6 + 2 \cos \alpha & \cos \alpha \\ 14 & 7 + 2 \sin \alpha & \sin \alpha \end{vmatrix}$, if $\alpha = 15^\circ$.

Select one:

- ☐ $-\frac{\sqrt{3}}{2}$
- ☒ 0 ✓
- ☐ 1
- ☐ $-\frac{2}{\sqrt{3}}$

Your answer is correct.

The correct answer is: 0